Data Preprocessing Documentation

Project: Intern Performance Analytics & Insights Dashboard

Sprint 1 – Data Collection and Preparation

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1. Introduction

This project focuses on analyzing intern performance using a structured dataset. Before performing analysis, the dataset must be inspected and prepared to ensure it is accurate, complete, and suitable for further processing.The purpose of preprocessing is to verify the dataset quality and make sure the data is ready for analysis.

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2. Dataset Overview

Property	Description
Dataset Name	Intern Insight Dataset
File Format	CSV
Number of Rows	100
Number of Columns	25
Tools Used	Python, Pandas, Numpy

The dataset contains information related to interns such as personal details, internship information, task completion, and attendance.

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3. Data Loading

The dataset was loaded into the Python environment using **pandas**, and **NumPy** was used for supporting numerical operations during preprocessing.

The CSV file was imported into a **DataFrame** so that it could be easily inspected and processed.

This step ensured that the dataset was successfully read and available for analysis.

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4. Data Inspection

Initial inspection of the dataset was performed to understand its structure and content.

The following checks were carried out:

- Viewing the first few rows of the dataset
- Checking column names and data types
- Verifying the number of rows and columns

These checks helped confirm that the dataset was correctly structured.

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5. Missing Value Check

A check was performed to determine whether any values were missing from the dataset.

The dataset was examined for null or empty values in each column.

Result:
No missing values were found in the dataset.

This indicates that the dataset is complete and does not require handling of missing data.

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6. Duplicate Record Check

Duplicate records were checked to ensure that the dataset does not contain repeated rows.

Duplicate entries can affect analysis results and lead to incorrect insights.

Result:
No duplicate rows were detected in the dataset.

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7. Data Type Verification

The data types of all columns were reviewed to confirm that they are appropriate.

Observations:

- Numerical features are stored as integer or float values.
- Categorical features are stored as text values.

Correct data types are important for performing accurate analysis and visualizations.

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8. Summary Statistics

Descriptive statistics were generated for numerical columns to better understand the dataset.

This analysis helped identify:

- Average values
- Minimum and maximum values
- Distribution of numeric variables

These statistics provide an overview of the dataset and help identify potential anomalies.

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9. Conclusion

The preprocessing and validation steps confirmed that the dataset is clean and ready for further analysis.

Key observations:

- The dataset contains 100 rows and 25 columns.
- No missing values were detected.
- No duplicate records were found.

The dataset has been validated and cleaned, and is now ready for exploratory data analysis and visualization.